INCH-POUND
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SUPERSEDING
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3 July 1997

PERFORMANCE SPECIFICATION

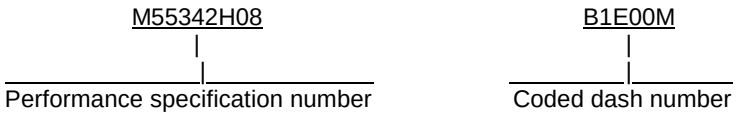
RESISTORS, FIXED, FILM, CHIP, NONESTABLISHED RELIABILITY, ESTABLISHED RELIABILITY, SPACE LEVEL, STYLE RM2010

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for style RM2010, fixed, film, chip, nonestablished reliability, established reliability, and space level resistors. This style is available in characteristic H, characteristic E, characteristic K, and characteristic M, resistance tolerances .1 percent, 1 percent, 2 percent, 5 percent, and 10 percent, and all termination materials except termination P.

1.2 Part or Identifying Number (PIN). Chip resistors covered by this specification must be identified by a PIN which must consist of the basic number of this specification and a coded dash number. The PIN must be in the following form:



The coded dash number must be derived in accordance with MIL-PRF-55342.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements documents cited in sections 3 and 4 of this specification, whether or not they are listed.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: US Army Communications - Electronics Command, ATTN: AMSEL-LC-LEO-E-EP, Fort Monmouth, NJ 07703-5023, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of the basic document or by letter.

AMSC N/A FSC 5905
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2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

DEPARTMENT OF DEFENSE

MIL-PRF-55342 - Resistors, Fixed, Film, Chip, Nonestablished Reliability, Established Reliability, Space Level, General Specification for.

(Unless otherwise indicated, copies of the above specifications, standards, and handbooks are available from the Document Automation and Production Service (DAPS), Building 4D (Customer Service), 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 General. The requirements for acquiring the product described herein shall consist of this document and MIL-PRF-55342.

3.2 Interface and physical dimensions. Resistors shall meet the interface and physical dimensions specified on figure 1, as applicable.

3.3. Power rating. The power ratings in milliwatts for characteristic H, characteristic E, characteristic K, and characteristic M relative to the associated test board shall be as follows:

Characteristic	Fiber board	Ceramic board
H and E	400 mW	400 mW
K and M	600 mW	800 mW

* 3.3.1 Power rating (power conditioning test). The power conditioning test for space level parts shall be based on the power rating of fiber boards (see 3.3).

3.4 Voltage rating. The maximum continuous working voltage shall not exceed 150 volts.

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3.5 Resistance and resistance tolerance. Minimum and maximum resistance values and associated resistance tolerances shall be as listed in table I.

* TABLE I. Minimum and maximum resistance values and tolerances.

Resistance tolerance	Minimum resistance	Maximum resistance
percent (±)		
0.1	100 ohms	4.99 megohms
1.0	5.62 ohms	15 megohms
2.0	5.6 ohms	15 megohms
5.0	5.6 ohms	15 megohms
10.0	5.6 ohms	15 megohms

4. VERIFICATION

4.1 Sampling and inspection. Sampling and inspection procedures shall be in accordance with MIL-PRF-55342.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order. When actual packaging of materiel is to be performed by DoD personnel, these personnel need to contact the responsible packaging activity to ascertain requisite packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activity within the Military Department or Defense Agency, or within the Military Department's System Command. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Chip resistors are intended to be used in thin or thick film hybrid circuits where micro-circuitry is indicated and in surface mount applications.

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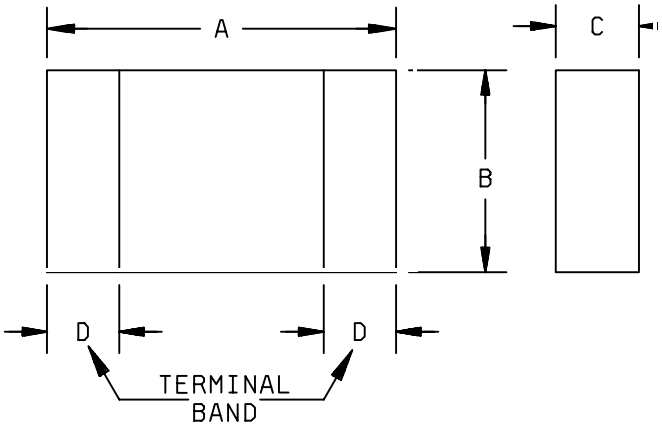
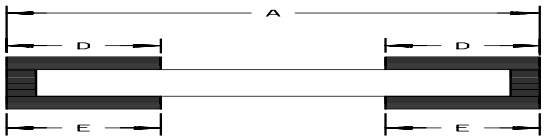
Configuration A



Configuration B



.020	0.51
.030	0.76
.033	0.84
.098	2.49
.206	5.23
.209	5.31

Configuration	Dimension A inch	Dimension B inch	Dimension C inch	Dimension D inch	Dimension E inch
A	.206 ±.005	.098 ±.005	.012/.030	.015 ±.005	N/A
B	.209 ±.009	.098 ±.005	.015/.033	.020 ±.005	.020 ±.005

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ?.005 (0.13 mm).
4. The pictorial view of the styles above are given as representative of the envelope of the item. Slight deviations from the outline shown, which are contained within the envelope, and do not alter the functional aspects of the device are acceptable.
5. Configuration A covers termination materials D, T, and W.
6. Configuration B covers termination materials B, C, G, and U.

FIGURE 1. Style RM2010.

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* 6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, date of this specification, the applicable associated specification, and the complete PIN (see 1.2).

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.2.1).

c. Packaging requirements (e.g., Electrostatic discharge (ESD) sensitivity) (see 5.1).

d. Allowable substitution (see MIL-PRF-55342).

e. If marking is required (see MIL-PRF-55342).

6.3 Tolerance for wraparound termination. The added tolerance for the wraparound type termination is intended to apply only to termination, metallization, and pretinning material.

* 6.4 Electrostatic charge effects. Under relatively low humidity conditions, some types of film resistors, particularly those with small dimensions and high sheet resistivity materials, are prone to sudden significant changes in resistance (usually reductions in value) and to changes in temperature coefficient of resistance as a result of discharge of static charges built up on associated objects during handling, packaging, or shipping. Substitution of more suitable implements and materials can help minimize this problem. For example, use of cotton gloves, static eliminator devices, air humidifiers, and operator and workbench grounding systems can reduce static buildup during handling. Means of alleviating static problems during shipment include elimination of loose packaging of resistors and use of metal foil (conductive) and static dissipation packaging materials. Direct shipments to the government is controlled by MIL-DTL-39032 which specifies a preventive packaging procedure."

* 6.5 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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Navy - AS, CG, MC, OS
Air Force - 19, 99

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